

Global Valuation

Cross-Country Valuation Multiples: Data and First Results

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Outline

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Data and Definitions

Data Sources

File	Size	Rows	Cols	Description
compustat_global	2.3 GB	981K	468	Global annual fundamentals (122 countries)
compustat_america	0.8 GB	254K	984	North America fundamentals
returns_global	1.9 GB	10.4M	25	Global monthly security prices

- **Compustat Global:** fiscal years 1987–2024, 122 countries
- **Compustat America:** fiscal years 1986–2024
- **Combined panel:** 743,087 firm-years, 72,720 firms, 81 countries
- All financial data in *local currency* (millions)

Variable Definitions

Dependent variables:

$$\log(P/E)_{i,t} = \log\left(\frac{\text{Market Cap}_{i,t}}{\text{Net Income}_{i,t}}\right), \quad NI > 0$$

$$\log(M/B)_{i,t} = \log\left(\frac{\text{Market Cap}_{i,t}}{\text{Book Equity}_{i,t}}\right)$$

Market Cap	US: mkvalt. Global: cshoi $\times$ prccm (primary security)
Book Equity	ceq (common equity). <i>Does not include debt.</i>
Net Income	ib (income before extraordinary items)
Leverage	(at – ceq)/at = non-equity share of assets
ROA	ib / at. Industry: GICS Industry Group (~25 categories)

All ratios winsorized at 1st/99th percentiles. All data in local currency (millions).

Market Cap Construction

US firms: `mkvalt` (pre-computed market value in USD millions). Available from 1986.

Non-US firms (Compustat Global): no pre-computed market value.

- Compute: $\text{Market Cap} = \text{cshoi} \text{ (shares outstanding)} \times \text{prccm} \text{ (month-end price)}$
- **Pitfall:** firms have multiple securities across exchanges and currencies

Example — SAP SE (gvkey=103487) in Dec 2022:

iid=02W EUR 96.39 ← correct

iid=05W HUF 38,620 ← wrong currency!

iid=06W MXN 2,009 ← wrong currency!

- **Solution:** use `prirow` field (primary security designation) to select the correct iid for each firm

All data in local currency (millions). Cross-country *levels* absorbed by country FE.

Accounting Standards

Compustat Global reports accounting standard via acctstd:

Standard	Share
IFRS	59.7%
Domestic GAAP	39.3%
US GAAP	0.2%
Other / missing	0.8%

US firms: all US GAAP.

Country variation:

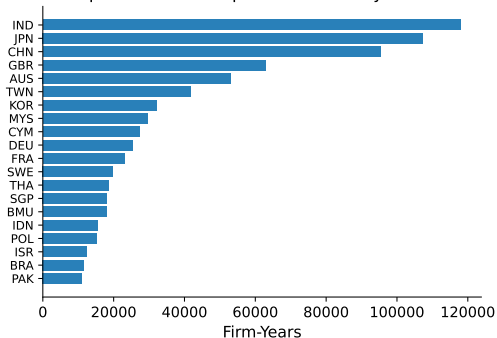
- EU: mandatory IFRS since 2005
- Japan: 93% domestic GAAP
- China: 96% domestic GAAP
- Korea: 99% IFRS (since 2011)
- India: 81% IFRS (since 2016)

Book equity (ceq) particularly affected: IFRS vs. local GAAP differ on intangibles and fair value.

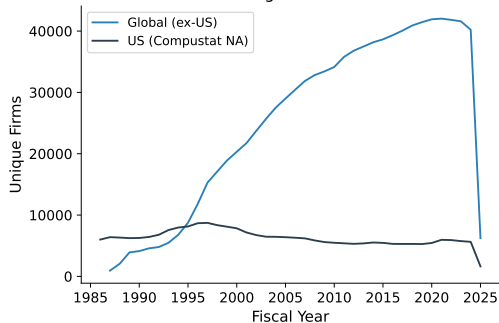
Country FE absorb time-invariant accounting differences. Concern: within-country switches (EU 2005).

Coverage: Countries and Time

Compustat Global: Top 20 Countries by Firm-Years



Coverage Over Time



India, Japan, China dominate non-US coverage. Panel peaks at ~41K firms (2022). Global price data starts 2007, so Global M/B and P/E limited to 2007–2024 (US from 1986).

Descriptive Statistics

Variable	N	Mean	P10	Median	P90
log(P/E)	531,558	2.94	1.76	2.79	4.32
log(M/B)	743,087	0.44	-0.78	0.38	1.73
Leverage	743,087	0.49	0.14	0.49	0.85
ROA	742,653	-0.01	-0.15	0.02	0.11
log(Assets)	743,087	7.28	3.27	7.17	11.48

Top 5 countries by obs:

USA (188K), JPN (60K), CHN (58K),
IND (57K), TWN (31K)

P/E restricted to profitable firms (72% of panel). Median P/E ≈ 16 , median M/B ≈ 1.5 .

Cross-Country Regressions

Regression Specifications

P/E (profitable firms, $NI > 0$):

$$\log(P/E)_{i,t} = \beta_1 \text{Lev} + \beta_2 \log(\text{Assets}) + \beta_3 \Delta E_{t-1} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

M/B (all firms):

$$\log(M/B)_{i,t} = \beta_1 \text{ROA} + \beta_2 \text{Lev} + \beta_3 \log(\text{Assets}) + \beta_4 \Delta E_{t-1} + \text{FE} + \varepsilon_{i,t}$$

ROA:

$$\text{ROA}_{i,t} = \beta_1 \text{Lev} + \beta_2 \log(\text{Assets}) + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

FE = γ_c (country) + δ_j (GICS industry) + τ_t (year). SE clustered at country level. Spec 2: $\gamma_{c,t}$ (country×year) replaces $\gamma_c + \tau_t$.

P/E Regression Results

	Spec 1	Spec 2
	C + Ind + Year FE	C×Year + Ind FE
Leverage	−0.017 (0.102)	−0.013 (0.100)
log(Assets)	−0.076** (0.030)	−0.078** (0.031)
Lag Earn. Growth	−0.016*** (0.003)	−0.016*** (0.003)
N / R^2	428,367 / 0.142	428,367 / 0.163

Profitable firms only. Leverage is insignificant; larger firms have lower P/E; past earnings growth *reduces* current P/E (mean reversion in earnings).

M/B Regression Results

	Spec 1	Spec 2
	C + Ind + Year FE	C×Year + Ind FE
ROA	0.259 (0.158)	0.225 (0.154)
Leverage	0.901*** (0.060)	0.902*** (0.063)
log(Assets)	−0.074*** (0.027)	−0.077*** (0.028)
Lag Earn. Growth	0.008*** (0.002)	0.007*** (0.002)
N / R^2	592,856 / 0.219	592,856 / 0.246

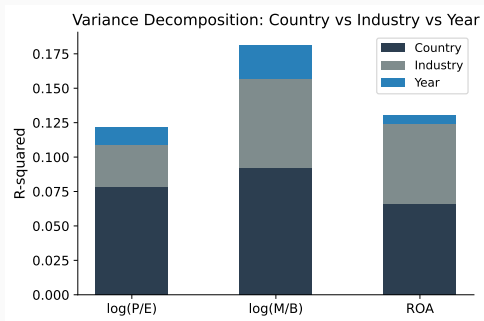
Leverage: +0.90*** (mechanical: less book equity $\Rightarrow$ higher M/B). ROA insignificant after controls.
M/B = Market Equity / Book Equity (not including debt).

P/E vs M/B: Key Regression Differences

- **Leverage:** Insignificant for P/E (-0.02) but dominant for M/B ($+0.90^{***}$)
 - M/B: less book equity $\Rightarrow$ higher market/book (mechanical)
 - P/E: price and earnings both scale with equity $\Rightarrow$ no effect
- **Earnings growth:** Negative for P/E (-0.016^{***}), positive for M/B ($+0.008^{***}$)
 - P/E: past earnings surge raises denominator $\Rightarrow$ lower P/E
 - M/B: past growth signals quality $\Rightarrow$ higher market premium
- **ROA:** Not included in P/E spec; insignificant in M/B after controls — current profitability does not predict cross-sectional valuation
- R^2 : P/E (0.142) < M/B (0.219) — M/B partly driven by leverage mechanics

What Drives Valuation Differences?

Variance Decomposition: Country vs Industry vs Year



Shapley R^2 decomposition:

	P/E	M/B	ROA
Country	64.6%	50.8%	50.7%
Industry	24.8%	35.8%	44.8%
Year	10.6%	13.4%	4.5%

Country dominates all three. P/E shows *strongest* country dependence.

Industry matters more for M/B than P/E (partly via leverage norms that vary by industry).

Leverage Varies Widely Across Countries

Highest median leverage:

Portugal 0.72

Tunisia 0.72

Italy 0.66

Spain 0.63

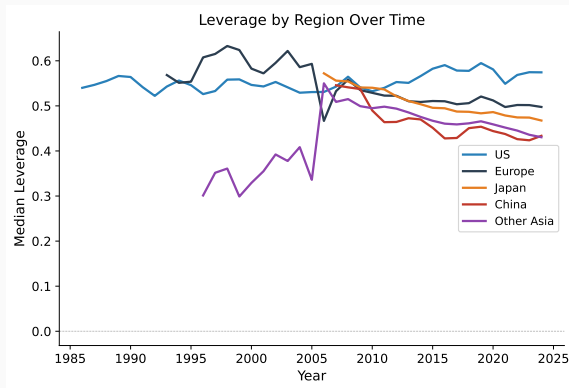
France 0.59

Australia 0.25

Lowest: Romania 0.24

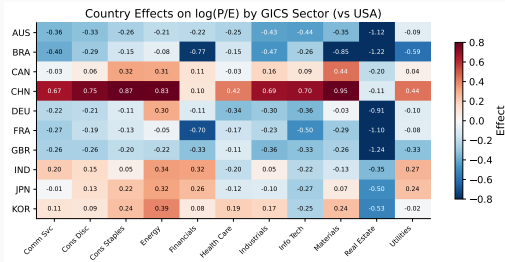
Guernsey 0.22

Cross-country std = 0.095

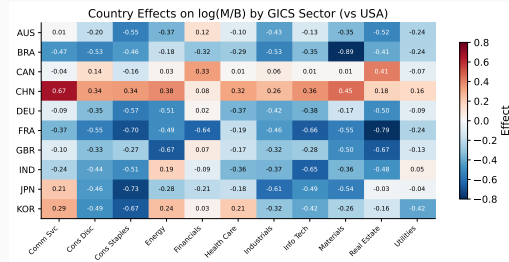


Southern Europe highly levered; Anglo-Saxon and Asia-Pacific lower. This heterogeneity makes M/B hard to compare across countries.

Industry $\times$ Country Interaction



P/E country effects by sector

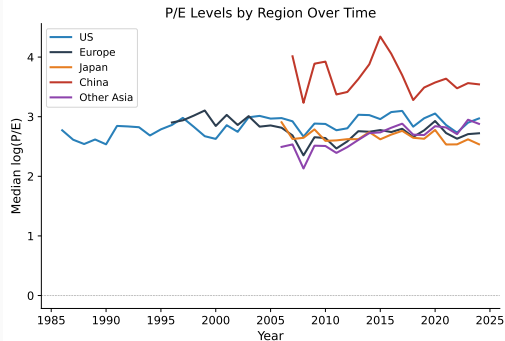


M/B country effects by sector

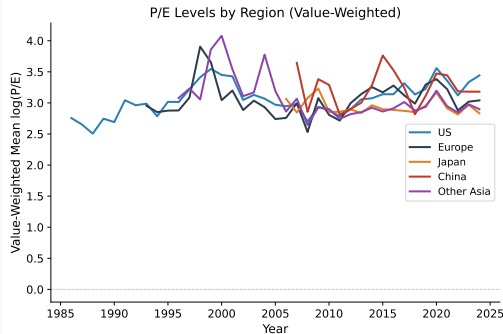
Country effects vs USA by GICS sector. Real Estate shows large negative effects across most countries in both metrics. China premium is broad-based across sectors.

Regional Trends

Regional P/E Over Time



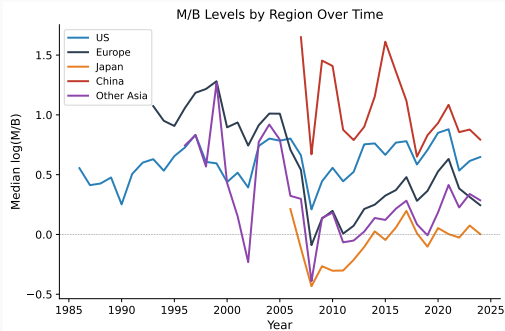
Median across firms



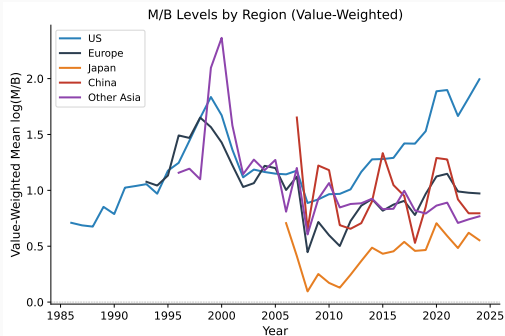
Value-weighted

China EW P/E is highest (many small firms with high P/E). VW shows US and China closer together. Japan P/E is comparable to Europe.

Regional M/B Over Time



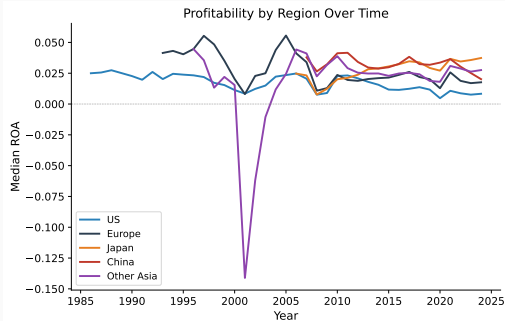
Median across firms



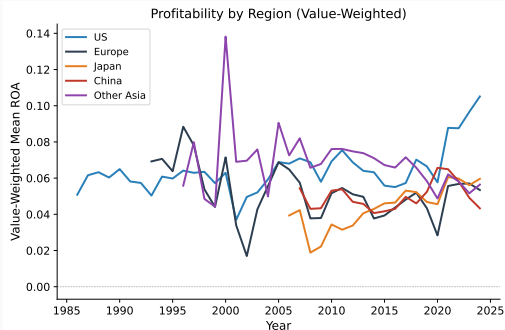
Value-weighted

Japan's deep M/B discount is stark, especially in VW. US VW M/B is much higher than any other region, driven by large tech firms.

Regional Profitability (ROA) Over Time



Median across firms



Value-weighted

Japan and Other Asia have *comparable or higher* profitability than US in EW, despite lower valuations.

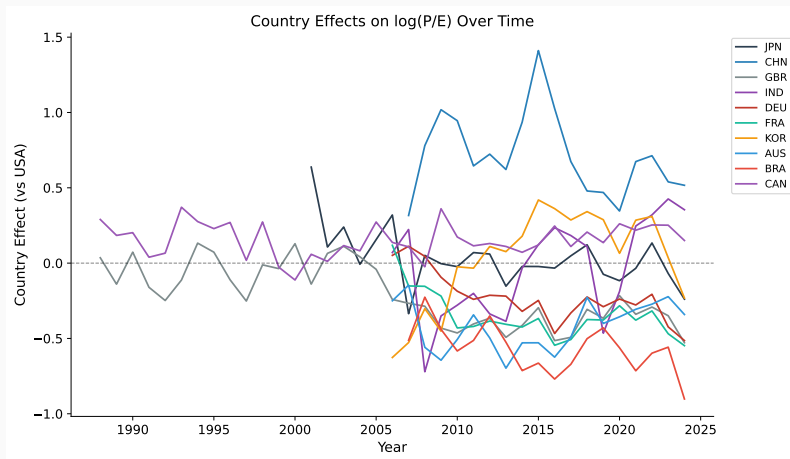
The US Valuation Premium (Recent 5 Years)

US minus	P/E gap		M/B gap	
	EW	VW	EW	VW
Europe	+18%	+24%	+33%	+123%
Japan	+36%	+54%	+98%	+258%
China	-48%	+7%	-18%	+132%
Other Asia	+7%	+48%	+50%	+190%

- VW gaps are much larger than EW — US premium concentrated in large firms (tech)
- Japan: P/E gap +54% but M/B gap +258% — much of the M/B gap is book equity measurement
- China: US—China EW gap is -48% (China P/E higher, driven by small firms); VW gap only +7%

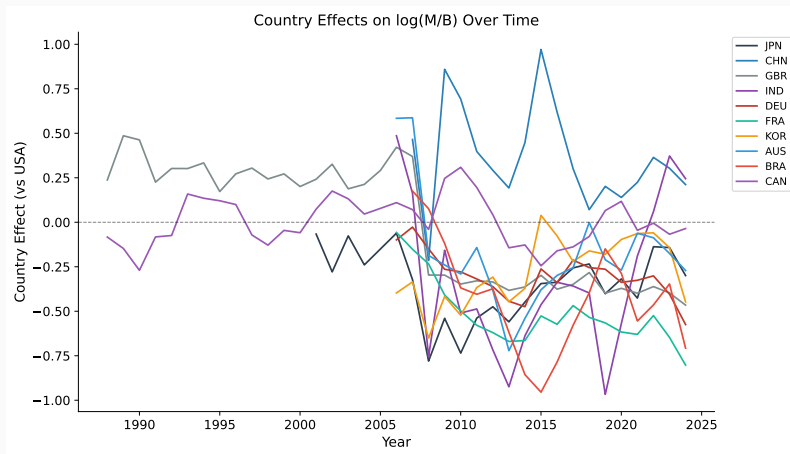
Country Effects Over Time

Country Effects on P/E (vs USA)



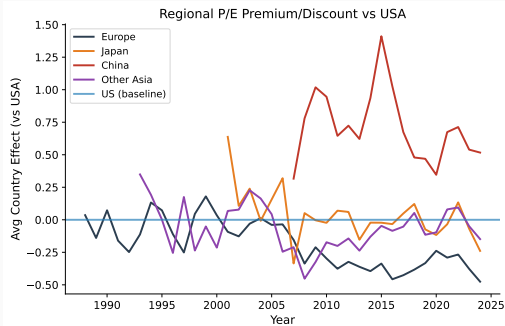
Year-by-year: $\log(P/E) \sim \text{Lev} + \log(\text{Assets}) + \Delta E_{t-1} + \text{Country FE} + \text{Industry FE}$. USA = 0.

Country Effects on M/B (vs USA)

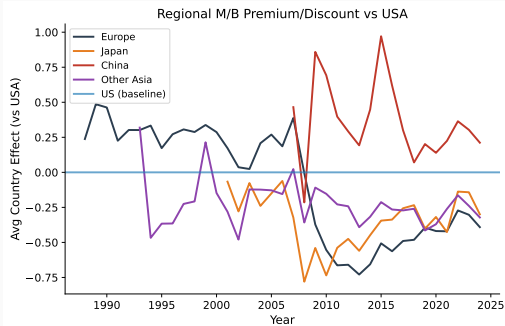


Year-by-year: $\log(M/B) \sim \text{ROA} + \text{Lev} + \log(\text{Assets}) + \Delta E_{t-1} + \text{Country FE} + \text{Industry FE}$. USA = 0.

Regional Valuation Gap vs USA



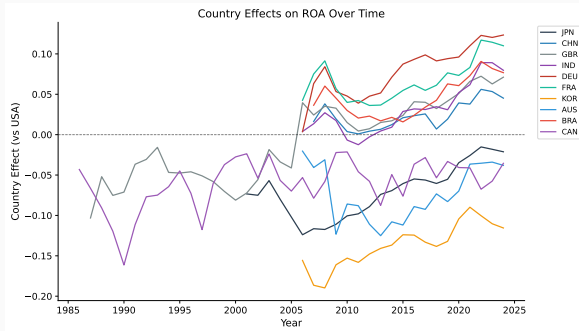
Average P/E country effects by region



Average M/B country effects by region

Both metrics: Europe and Japan below US, China above. The P/E-M/B gap for Japan (near zero in P/E, large discount in M/B) is the most striking difference.

Japan: The P/E vs M/B Puzzle



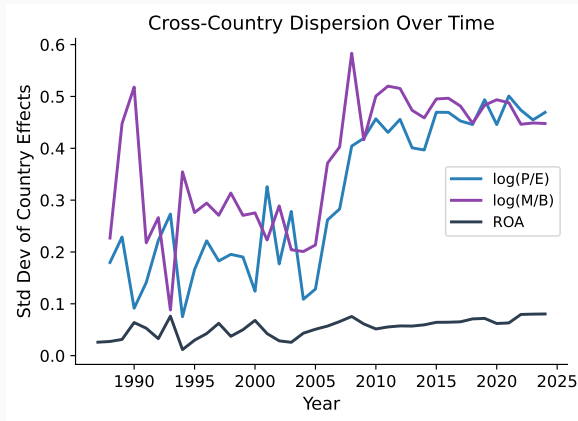
Japan's “discount” depends on the metric:

- M/B: -0.35 (negative 100% of years)
- P/E: $+0.03$ (near zero)
- ROA: positive (high profitability)

Japan's M/B discount reflects *conservative book values* (domestic GAAP, 93%), not lower market valuations vs earnings.

P/E is a cleaner comparison.

Cross-Country Dispersion Over Time



Dispersion is diverging:

- P/E: 0.17 $\rightarrow$ 0.47
- M/B: 0.40 $\rightarrow$ 0.45

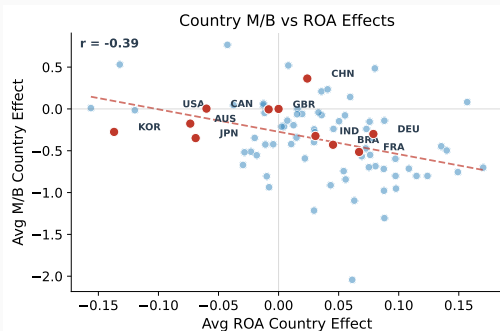
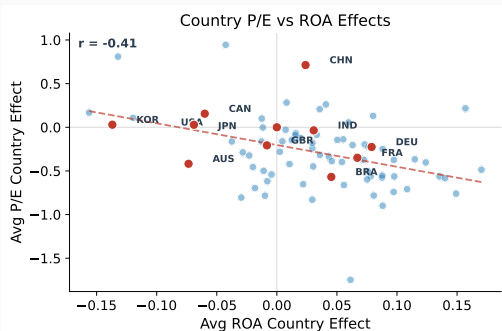
P/E dispersion nearly tripled ($\times 2.7$).
M/B dispersion was more stable.

ROA dispersion more stable
(~ 0.05 – 0.10), with a spike during
the GFC (2009–10).

Valuations are *diverging* despite
financial globalization.

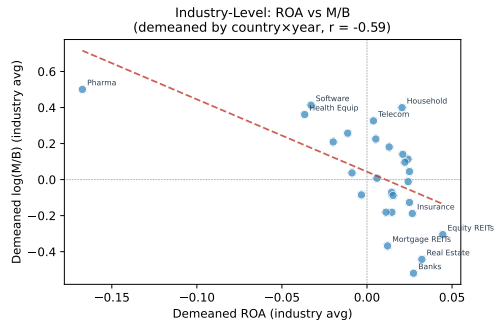
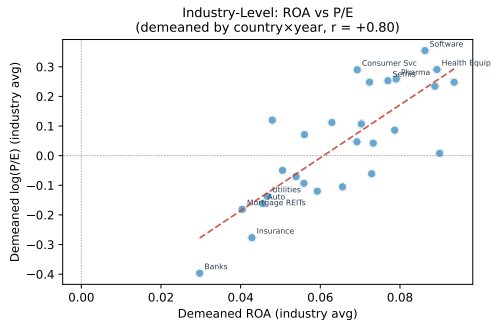
Valuation, Profitability, and Growth

High-Valuation Countries Are Not More Profitable



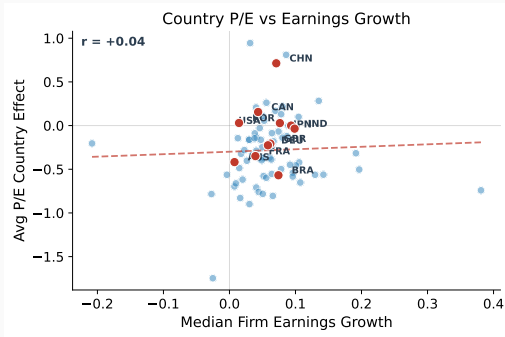
Each dot is a country (average effect vs USA across years). The negative correlation ($r = -0.41$ for P/E, $r = -0.38$ for M/B) means high-valuation countries tend to have *lower* profitability. Brazil: deeply discounted yet profitable. China: high P/E but average ROA.

Within Countries, the ROA–Valuation Link Differs by Metric



Each dot is a GICS industry group (27 industries, demeaned by country $\times$ year). **P/E** ($r = +0.80$): within a country, more profitable industries get higher P/E. Software and Health Equipment at top; Banks and Insurance at bottom. **M/B** ($r = -0.59$): *reverses*. Pharma has lowest ROA but highest M/B (growth options, intangibles). Banks: high ROA, lowest M/B.

Earnings Growth Does Not Explain Valuations Either



Cross-country correlations:

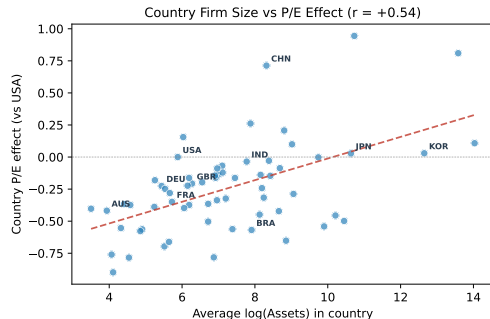
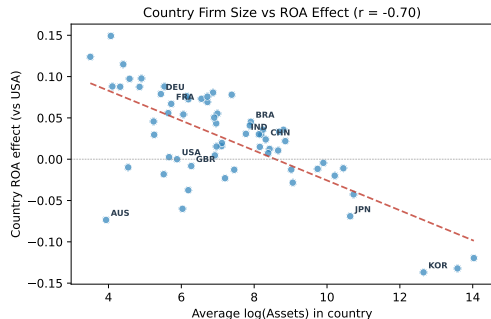
	P/E	M/B
vs ROA	-0.41	-0.38
vs Earn. Growth	-0.09	+0.03

Earnings growth is *uncorrelated* with valuations. Japan has higher growth than the US yet near-zero P/E premium. Brazil has US-level growth yet a deep discount.

Implication: the cross-country valuation spread is a *discount rate* phenomenon, not a cash flow story.

What Explains the ROA–Valuation Puzzle?

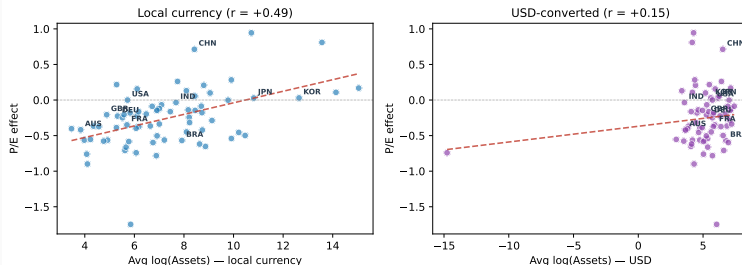
Firm Size in Local Currency Absorbs the Puzzle...



Each dot is a country. Size = average $\log(\text{Assets})$ in *local currency*. Countries with smaller firms have higher ROA ($r = -0.70$) and lower P/E ($r = +0.54$). Controlling for size renders ROA insignificant on P/E ($t: -4.57 \rightarrow -1.63$). But is size real or a currency artifact?

...But the Size Effect Is a Currency Artifact

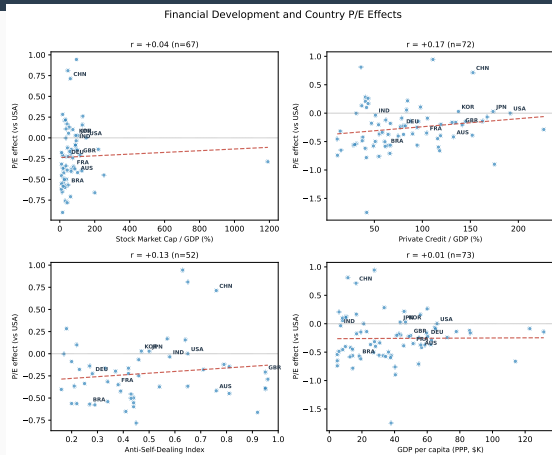
Does Currency Conversion Change the Size-P/E Relationship?



	Size vs P/E	Size vs ROA	Partial $r(\text{ROA}, \text{P/E} \mid \text{size})$
Local currency	+0.49	-0.77	-0.17
USD-converted	+0.15	-0.12	-0.47

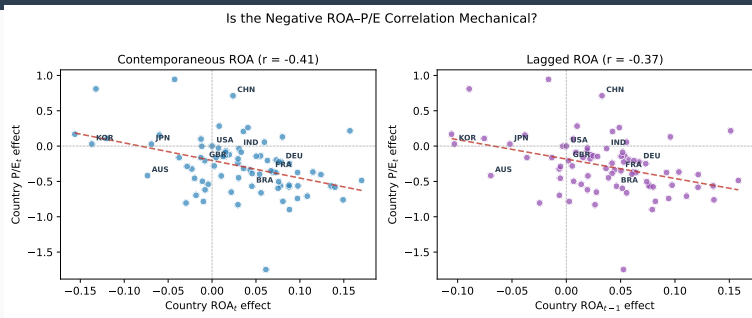
Local currency: JPN $\log(at) = 10.8$, KOR 12.6, IDN 13.6. In USD: 6.1, 5.6, 4.2. After conversion, size no longer absorbs the ROA-P/E link.

Standard Financial Development Indicators Do Not Explain P/E



Market cap/GDP ($r = +0.04$), credit/GDP ($r = +0.17$), anti-self-dealing ($r = +0.13$), GDP per capita ($r = +0.01$): all near zero.

The Correlation Is Not Mechanical (Lagged ROA Test)



Concern: NI is in both ROA (numerator) and P/E (denominator) — a temporary earnings shock mechanically raises ROA and lowers P/E.

Test: replace ROA_t with ROA_{t-1} . Last year's NI is not in this year's P/E.

Result: correlation barely changes ($-0.41 \rightarrow -0.37$). Country ROA effects are extremely persistent ($r = 0.99$), so the link reflects *permanent* profitability differences, not transitory earnings.

Nothing Absorbs the ROA–P/E Link

Control variable	Corr with		Partial r (ROA, P/E control)	N
	P/E	ROA		
<i>None (raw)</i>	—	—	−0.41	81
Stock Market Cap / GDP	+0.04	−0.17	−0.50	67
Private Credit / GDP	+0.17	−0.20	−0.45	72
Anti-Self-Dealing Index	+0.13	−0.24	−0.55	52
Rule of Law	+0.05	−0.03	—	73
GDP per capita (PPP)	+0.01	+0.01	—	73
Avg firm size (local ccy)	+0.49	−0.77	−0.17	72
Avg firm size (USD)	+0.15	−0.12	−0.47	72

Standard institutional variables have near-zero correlation with P/E effects. Local-currency size appeared to absorb the puzzle, but was a **currency artifact**. The ROA–P/E puzzle remains **unexplained**.

Stylized Facts and Takeaways

Are P/E and M/B Results Aligned?

Broadly yes (country-level correlation = +0.85), **but with key differences:**

Finding	P/E	M/B
Country share of R^2	64.6%	50.8%
Industry share of R^2	24.8%	35.8%
Leverage coefficient	-0.02 (ns)	+0.90***
Japan effect (avg)	+0.03	-0.35
Dispersion trend	$\times 2.7$	$\times 1.1$

Key insight: M/B is contaminated by leverage norms and book equity accounting (IFRS vs domestic GAAP). P/E reveals even *stronger* country dependence.

Stylized Facts (1/2): Cross-Sectional Structure

1. **Country effects dominate valuation** — country explains 51–65% of R^2 (Shapley), more in P/E than M/B. Extends Heston–Rouwenhorst (1994) from *returns* to *levels*.
2. **Country dominates profitability too** — for ROA, country edges industry (51% vs 45%).
3. **Negative valuation–ROA correlation across countries** ($\rho \approx -0.4$) — *not* explained by financial development (La Porta), GDP per capita, or firm size (in USD). Within countries, P/E–ROA flips positive (+0.80). The cross-country puzzle remains open.
4. **M/B is leverage-contaminated** — leverage coefficient +0.90*** in M/B, insignificant in P/E. Cross-country leverage heterogeneity (std = 0.095) mechanically inflates M/B variation.

Stylized Facts (2/2): Dynamics

5. **Japan discount is mostly a measurement artifact** — M/B effect -0.35 (100% negative years), but P/E effect ≈ 0 . Conservative Japanese book equity drives the M/B gap.
6. **China premium is real but declining** — high in both P/E and M/B, but falling since 2015.
7. **Valuation dispersion is diverging** — P/E dispersion nearly tripled. Despite financial globalization, country valuations are *not converging*.
8. **US VW premium is enormous** — driven by large cap: $+54\%$ P/E and $+258\%$ M/B vs Japan (VW). The US premium is much smaller in EW, suggesting it is concentrated in large tech firms.

Literature

Related Literature

Country vs industry effects:

- Heston–Rouwenhorst (1994, *JFE*) — country dominates returns. We extend to *levels*.
- Griffin–Karolyi (1998, *JFE*) — industry effects vary across sectors.

Institutions and investor protection:

- La Porta et al. (2002, *JF*) — investor protection raises Tobin's Q.
- Hail–Leuz (2006, *JAR*) — institutions reduce cost of equity by up to 200 bps.
- Stulz (2005, *JF*) — “twin agency problems” prevent equalization.

Market-to-book and accounting:

- Penman et al. (2007, *JAR*) — decompose M/B into leverage and operating components.
- Our P/E vs M/B divergence on Japan echoes this concern about M/B.

Research Agenda

Research Questions

1. **Why is P/E dispersion diverging?** ($\times 2.7$) — US tech sector? Post-GFC monetary policy? Institutional changes?
2. **What explains China's P/E premium?** — Highest P/E despite moderate M/B. Retail investors? Capital controls? Growth expectations?
3. **Can we decompose the country effect?** — Discount rates, growth expectations, investor protection, capital controls, accounting norms
4. **Why are valuations negatively correlated with ROA?** ($\rho \approx -0.4$) — Not explained by financial development, investor protection, GDP per capita, or firm size (in USD). Local-currency size appeared to work but was a currency artifact. A genuine open puzzle.
5. **Japan as natural experiment** — $P/E \approx 0$ vs $M/B = -0.35$: can we bound the “accounting noise” in M/B?

Suggested Next Steps

1. **EV/EBITDA** as third metric — avoids book equity issues (M/B) and profitable-firms restriction (P/E)
2. **Institutional variables** — La Porta variables surprisingly weak; try capital controls, FDI openness, expected growth proxies, risk measures
3. **Realized returns** — do low-valuation countries earn higher returns? (Bekaert–Harvey, Fama–French international)
4. **Persistence** — Markov transition matrices for country valuation quintiles
5. **Sector decomposition** — tradable vs non-tradable; Real Estate shows extreme country effects
6. **Within-country dispersion** — does valuation heterogeneity predict aggregate returns?